EBS VOLUME IN EC2 (LINUX)

An EBS volume is a block storage device used with Amazon Web Services EC2 instances. It provides persistent storage that can be attached to a virtual server. EBS volumes work like a hard disk for an EC2 instance. Data stored in the volume remains safe even if the instance is stopped.

Features of EBS Volume

- Provides persistent block storage for EC2 instances
- Data remains available even after instance restart
- Dependent service
- Can be attached and detached from instances
- Can create snapshots for backup

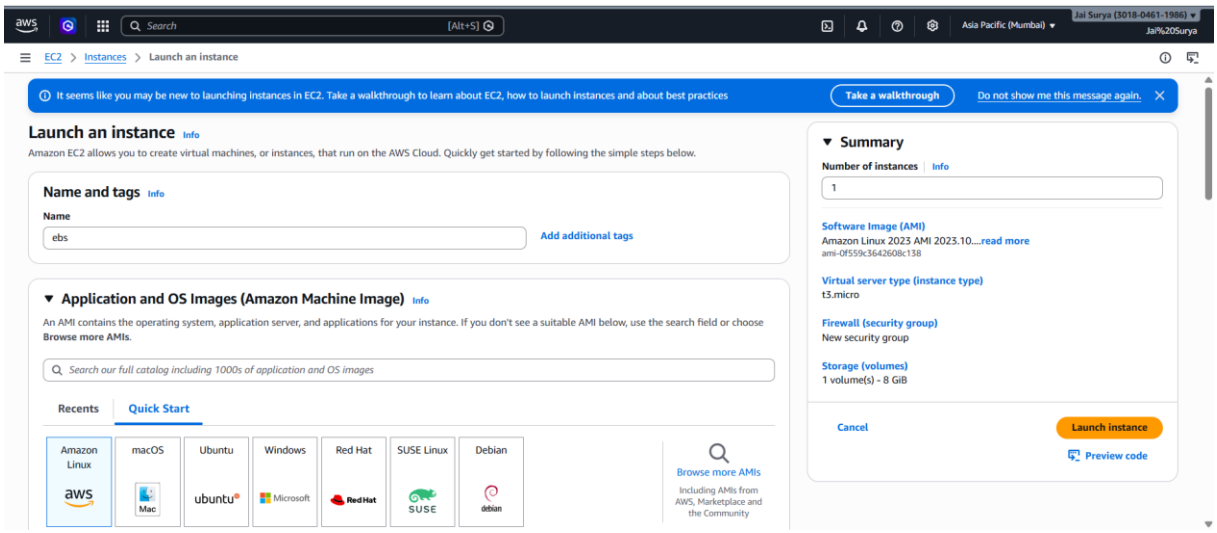
Types of EBS Volumes

- General Purpose SSD (gp2 / gp3) – Balanced price and performance
- Provisioned IOPS SSD (io1 / io2) – High performance applications
- Throughput Optimized HDD (st1) – Big data and large workloads
- Cold HDD (sc1) – Low-cost storage for infrequent access

HANDS-ON: EBS VOLUME IN EC2 (LINUX)

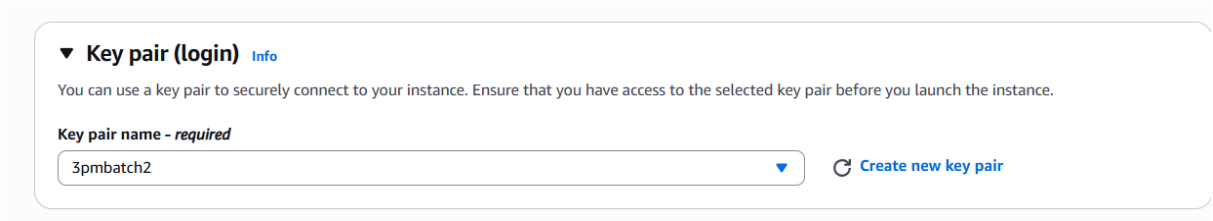
Screenshot: Creating Linux EC2 Instance

- Entered the instance name
- Selected Linux operating system



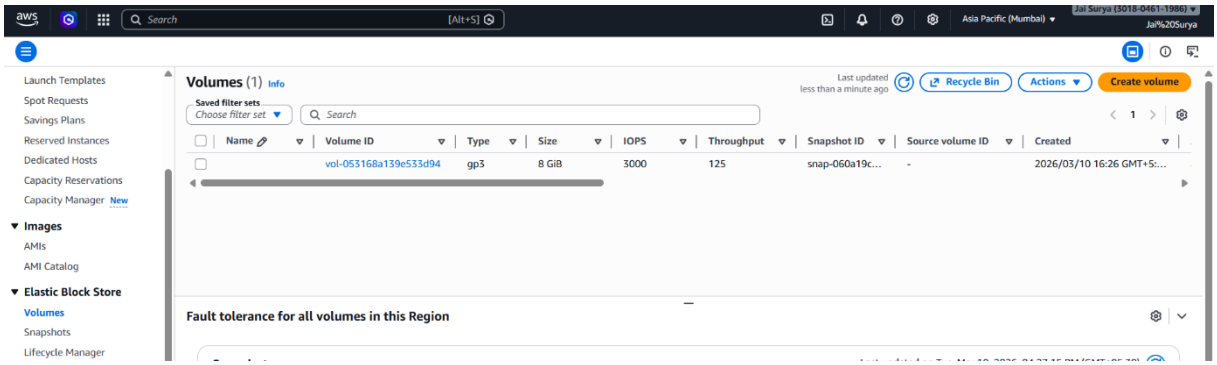
Screenshot: Selecting Key Pair

- Selected an existing key pair or created a new one



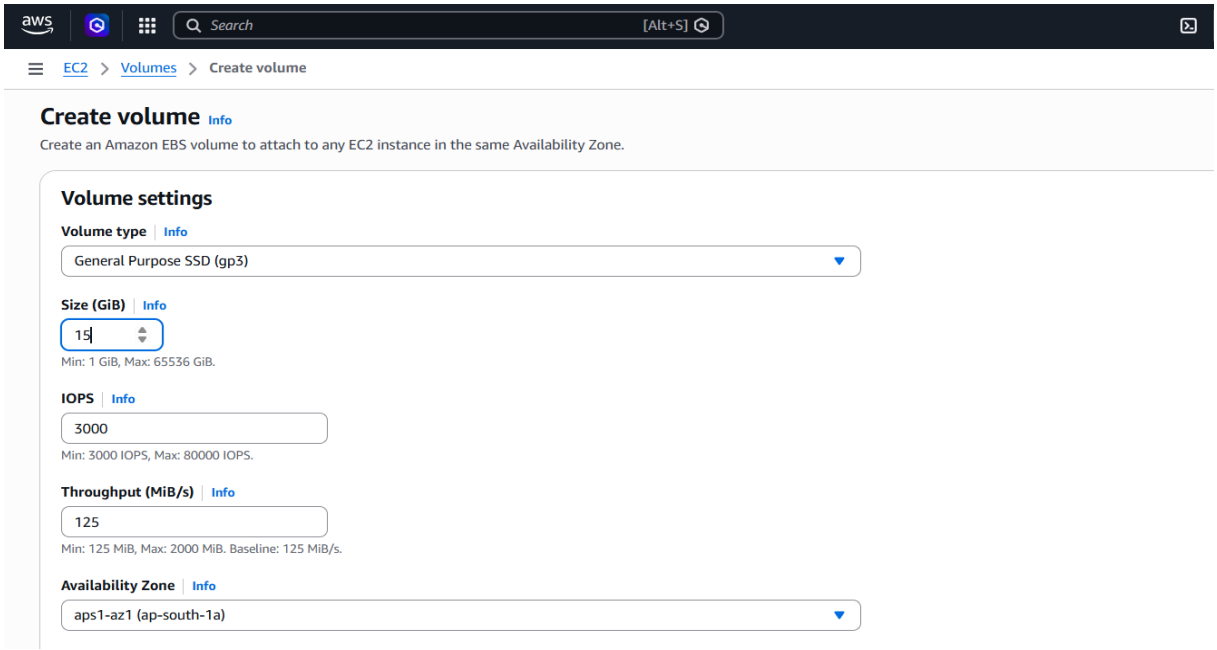
Screenshot: Creating EBS Volume

- Navigated to Elastic Block Store → Volumes
- Clicked Create Volume



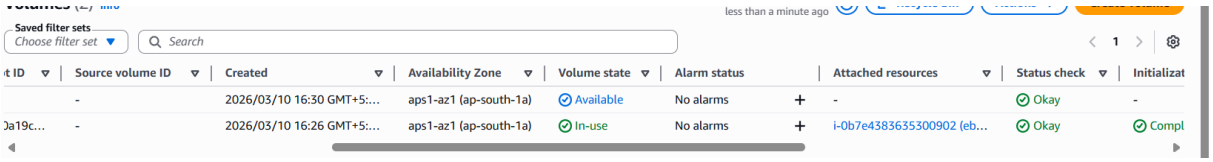
Screenshot: Configuring EBS Volume

- Selected the volume size
- Chose the same availability zone as the EC2 instance



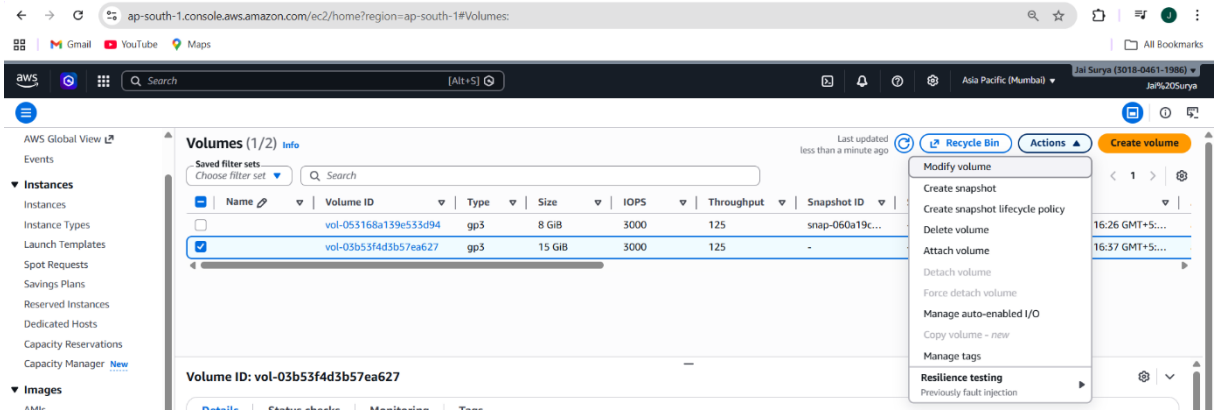
Screenshot: EBS Volume Created

- EBS volume is created successfully
- Volume state shows Available



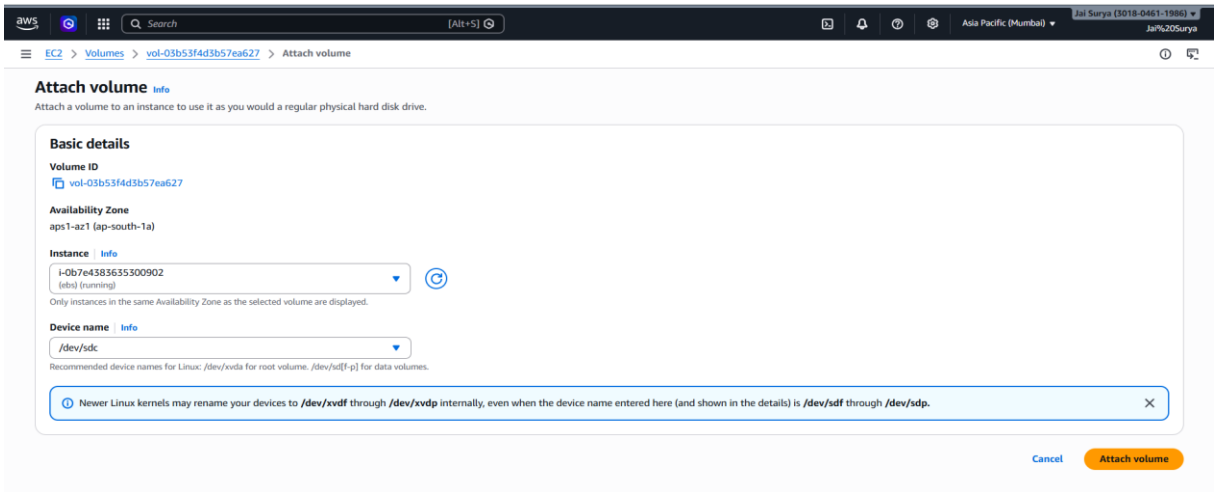
Screenshot: Attaching EBS Volume

- Selected the created EBS volume
- Clicked Actions
- Selected Attach Volume to connect it to the EC2 instance



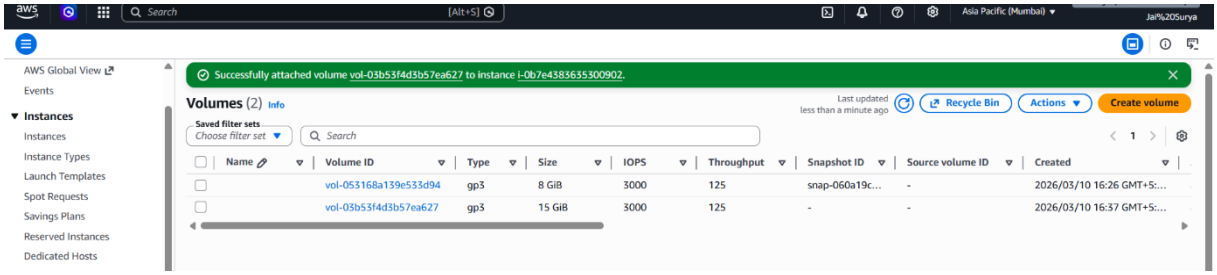
Screenshot: Selecting Instance and Device Name

- Selected the EC2 instance to attach the volume
- Entered the device name
- Proceeded to attach the EBS volume to the instance



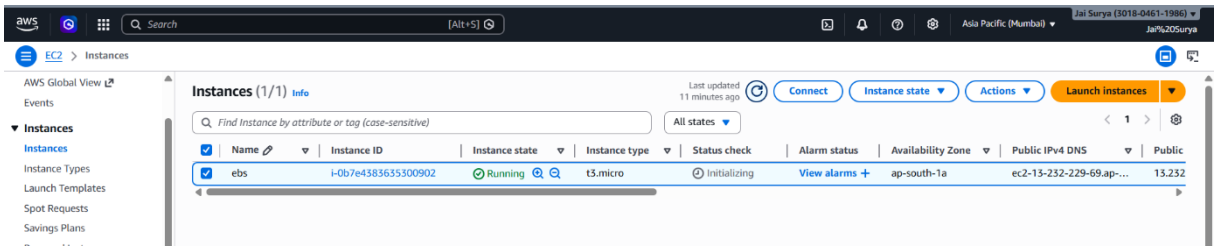
Screenshot: EBS Volume Attached

- EBS volume is successfully attached



Screenshot: Connecting to EC2 Instance

- Selected the EC2 instance
- Clicked Connect



Screenshot: Checking Attached Volumes

- Executed the command to view attached storage devices

  - lsblk

- Displayed the EBS volume attached to the EC2 instance.

```
[ec2-user@ip-172-31-47-190 ~]$ lsblk
NAME          MAJ:MIN RM SIZE RO TYPE MOUNTPOINTS
nvme0n1       259:0    0   8G  0 disk
├─nvme0n1p1   259:1    0   8G  0 part /
├─nvme0n1p127 259:2    0   1M  0 part
└─nvme0n1p128 259:3    0  10M  0 part /boot/efi
nvme1n1       259:4    0  15G  0 disk
```

Screenshot: Checking Disk Usage

- Executed the command to check disk usage

  - **df -h**

- The new EBS volume is not shown because it is not formatted yet.

```
nvme1n1       259:4    0   15G  0 disk
[ec2-user@ip-172-31-47-190 ~]$ df -h
Filesystem        Size  Used Avail Use% Mounted on
devtmpfs          4.0M     0  4.0M   0% /dev
tmpfs             459M     0  459M   0% /dev/shm
tmpfs             184M  440K  183M   1% /run
/dev/nvme0n1p1    8.0G  1.6G  6.4G  20% /
tmpfs             459M     0  459M   0% /tmp
/dev/nvme0n1p128   10M  1.3M  8.7M  13% /boot/efi
tmpfs              92M     0   92M   0% /run/user/1000
[ec2-user@ip-172-31-47-190 ~]$
```

Screenshot: Creating Mount Directory and Checking File System

- Switched to root user.

  sudo su

- Created a directory for mounting the EBS volume.

  mkdir /mnt/jai

- Checked the file system of the attached volume.

  file -s /dev/nvme1n1

```
[ec2-user@ip-172-31-47-190 ~]$
[ec2-user@ip-172-31-47-190 ~]$ sudo su
[root@ip-172-31-47-190 ec2-user]# mkdir /mnt/jai
[root@ip-172-31-47-190 ec2-user]# file -s /dev/nvme1n1
/dev/nvme1n1: data
[root@ip-172-31-47-190 ec2-user]#
```

Screenshot: Creating File System for EBS Volume

- Created a file system on the attached EBS volume.

  mkfs -t xfs /dev/nvme1n1

```
/dev/nvme1n1: data
[root@ip-172-31-47-190 ec2-user]# mkfs -t xfs /dev/nvme1n1
meta-data=/dev/nvme1n1           isize=512    agcount=16, agsize=245760 blks
         =                       sectsz=512   attr=2, projid32bit=1
         =                       crc=1        finobt=1, sparse=1, rmapbt=0
         =                       reflink=1    bigtime=1 inobtcount=1 nrext64=0
         =                       exchange=0
data     =                       bsize=4096   blocks=3932160, imaxpct=25
         =                       sunit=1      swidth=1 blks
naming   =version 2              bsize=4096   ascii-ci=0, ftype=1, parent=0
log      =internal log           bsize=4096   blocks=16384, version=2
         =                       sectsz=512   sunit=1 blks, lazy-count=1
realtime =none                   extsz=4096   blocks=0, rtextents=0
```

Screenshot: Mounting the EBS Volume And Verifying Mounted EBS Volume

- Mounted the EBS volume to the created directory.

  mount /dev/nvme1n1 /mnt/jai

- Checked the attached disks.

  lsblk

- Checked the mounted storage and disk usage.

  df -h

- The EBS volume is now visible and mounted successfully.

```
[root@ip-172-31-47-190 ec2-user]# mount /dev/nvme1n1 /mnt/jai
[root@ip-172-31-47-190 ec2-user]# lsblk
NAME          MAJ:MIN RM SIZE RO TYPE MOUNTPOINTS
nvme0n1       259:0    0   8G  0 disk
├─nvme0n1p1   259:1    0   8G  0 part /
├─nvme0n1p127 259:2    0   1M  0 part
└─nvme0n1p128 259:3    0  10M  0 part /boot/efi
nvme1n1       259:4    0  15G  0 disk /mnt/jai
[root@ip-172-31-47-190 ec2-user]# df -h
Filesystem        Size  Used Avail Use% Mounted on
devtmpfs          4.0M     0  4.0M   0% /dev
tmpfs             459M     0  459M   0% /dev/shm
tmpfs             184M  440K  183M   1% /run
/dev/nvme0n1p1    8.0G  1.6G  6.4G  20% /
tmpfs             459M     0  459M   0% /tmp
/dev/nvme0n1p128   10M  1.3M  8.7M  13% /boot/efi
tmpfs              92M     0   92M   0% /run/user/1000
/dev/nvme1n1       15G  140M   15G   1% /mnt/jai
[root@ip-172-31-47-190 ec2-user]#
```

Screenshot: Viewing Command History

- Displayed the previously executed commands.

  history

```
root@ip-172-31-47-190 ec2-user]# history
   1  mkdir /mnt/jai
   2  file -s /dev/nvme1n1
   3  mkfs -t xfs /dev/nvme1n1
   4  mount /dev/nvme1n1/mnt/jai
   5  mount /dev/nvme1n1 /mnt/jai
   6  lsblk
   7  df -h
   8  history
```